

ELPRO Analytics Dashboard Modifications — 2026-04-22

OBJECTIVE:

Replace old Test Prep graphs with ELPRO-specific performance charts for Class 10.

CHANGES MADE:

1. REMOVED GRAPHS (5 charts):

- Average Test Prep Score by Class
- Overall Class Test Prep Score Trend
- Individual Student Test Prep — [Student Name]
- Student Correct / Incorrect / Skipped
- Time Spent vs Score

2. ADDED GRAPHS (4 charts):

A. SCHOOL-WIDE AVERAGES (no filters):

- Avg 1st Test vs Best Score by Subject (grouped bar)
Shows: 1st Test Score % vs Best Score % for Math & Science
- Avg 1st Test vs Best Time by Subject (grouped bar)
Shows: 1st Test Time (sec) vs Best Time (sec) for Math & Science

B. PER-STUDENT ANALYSIS (with filters):

- Score: 1st Test vs Best Score (grouped bar)
- Speed: 1st Test vs Best Time (grouped bar)

3. FILTERS ADDED:

- Class (dropdown, dynamically populated from API data)
- Section (cascading, filtered by selected Class)
- Student (cascading, filtered by Class + Section)

Layout: School-wide charts first (no filters) → Divider → Filters above student charts

4. API CHANGES:

- Reduced fetch limit from 500 to 100 items (performance improvement)
- Data source: `dashboardAPI.getTestPrepBySchoolCode()`

5. TIME CALCULATION LOGIC:

For each student per subject:

- First Time = time of their chronologically first attempt
- Best Time = minimum (fastest) time across all their attempts

Then averaged across all students per subject for display

6. CHART FEATURES:

- All charts include X-axis labels showing subject/student names
- Grouped bar colors:
 - * Score: light blue (#93C5FD) vs dark blue (#1E3A5F)
 - * Time: orange (#F97316) vs yellow (#EAB308)
- Loading skeletons for async data
- Error handling for API failures

DATA STRUCTURE:

ElproStudentSummary interface:

- username: string
- className: string
- section: string
- subject: string (MATHEMATICS or SCIENCE)
- firstScore: number (%)
- bestScore: number (%)
- improvement: number (%)
- firstTime: number | null (seconds)
- bestTime: number | null (seconds, minimum across all attempts)
- attempts: number (count of attempts)

KNOWN ISSUES/NOTES:

1. Class field may not map correctly if API doesn't populate class_name
 - Section filter may show all sections regardless of class
 - Needs verification: check API response for class_name field structure
2. Some students show identical 1st vs Best times (e.g., 186/186 sec)
 - This is correct if student only attempted the subject once
 - Different values appear when students have multi-attempt data
3. Section B appearing for Class 10 when shouldn't exist
 - Suggests class_name field is not coming through correctly from API
 - **Needs debug:** inspect API response for actual class_name values

FILE LOCATIONS:

Component: src/components/AnalyticsSections.tsx

Charts: src/components/AnalyticsCharts.tsx (updated with showXAxisLabels prop)

Types: src/types/index.ts

API: src/services/api.ts